

Advanced Cisco Networking Academy - Multi- Protocol Route Redistribution with eBGP

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Purpose

Often in large networks, or when merging multiple networks together, a variety of routing protocols will be used. Route redistribution allows different routing protocols to communicate and share information, offering flexibility to choose the most suitable protocol for specific network segments. This lab outlines the process of setting up redistribution between multiple networks that operate using different routing protocols, and demonstrates how eBGP can be used for precise policy control and path manipulation.

Background

In the late 1900's, networking began to change dramatically. There was an outbreak of new routing protocols that redefined the optimal way to route traffic throughout a network. Many protocols came from this boom, but a few of them caught on, becoming the dominant protocols used to route traffic around the world.

Developed in the late 1980's through the 90's, Open Shortest Path First (OSPF) is an intra-domain link-state interior gateway protocol designed to route packets throughout a network using the shortest possible pathway from the source to the destination. Each router builds a map of the network using information learned during an exchange process where the routers share data about their directly connected links. From this data, the Shortest Path First (SPF) algorithm attempts to find a route with the smallest hop count to any given destination. Each router partaking in this protocol is given two types of identification: Router IDs and Process IDs. The Process ID points towards a specific instance of OSPF on the router, allowing the device to run multiple OSPF processes simultaneously. These IDs are seen only by the router they are assigned to, meaning they can be duplicated across the network. Router IDs serve as a unique identifier for the router itself, giving it a distinct identity within the network. This ID is public, so it must be unique.

In 1986, Interior Gateway Routing Protocol (IGRP) was designed and successfully deployed. As the name suggests, it is an Interior Gateway Protocol (IGP), yet it is frequently used as an Exterior Gateway Protocol (EGP) for inter-domain routing. IGRP calculates network routes using distance vector routing technology, which shares routing information based on the principle that each router does not need to know all network links. Instead, each router advertises destinations with a distance value, representing information about bandwidth, delay, load utilization, and link reliability. When a router receives this data, it adjusts the distance and propagates it to neighboring routers. Enhanced Interior Gateway Routing Protocol (EIGRP) utilizes the same distance vector technology, but it significantly improves convergence properties and overall efficiency. Using the Diffusing Update Algorithm (DUAL), only the routers impacted by a topology change will recalculate their data tables simultaneously, greatly speeding up the entire process.

Later, in 1992, Intermediate System – Intermediate System (IS-IS) was officially standardized by the International Organization for Standardization (ISO) after having been published by the Internet Engineering Task Force (IETF) in 1990. IS-IS is an IGP, link-state routing protocol that mirrors OSPF in many ways. Both protocols have areas, exchange link-state packets, and use the SPF algorithm to calculate pathways. However, unlike OSPF, IS-IS was developed for Connectionless-mode Network Protocol (CLNP), the more prominent network protocol of the time. Later, a team at Digital Equipment Corporation (DEC), including individuals such as Radia Perlman – the inventor of the Spanning Tree Protocol – modified IS-IS to route IP, renaming it integrated IS-IS. With this development, IS-IS was given a two-level hierarchy. Level 1 is used for intra-area routing whereas level 2 provides inter-area routing.

In 1989, a new routing protocol was developed that would quickly become the backbone of the Internet. Border Gateway Protocol (BGP) is a path-vector routing protocol that breaks a complex network into autonomous systems and advertises path information between these systems. On the scale of the Internet, autonomous systems are typically given to Internet Service Providers (ISPs) or large organizations. As of 2025, there are over 120,000 allocated autonomous system numbers on the internet.

With BGP, routes can be exchanged between autonomous systems using internal BGP (iBGP) and between autonomous systems using external BGP (eBGP). However, it is important to note that autonomous systems do not need to use iBGP to exchange routing information with another autonomous system via eBGP. BGP essentially acts like an international mail carrier, following specific guidelines to deliver mail from one country to another. Once the mail has reached the destination country, it will go through the local mail service to reach the recipient.

BGP has been widely adopted due to its security, scalability, and control. BGP authenticates messages between routers using preconfigured passwords, filtering out unauthorized traffic. Additionally, BGP is very efficient and can manage and adapt to large network topologies. Furthermore, BGP utilizes multihoming for fault tolerance, establishing backup circuits to prevent downtime in the event of a link failure. When determining these many routing paths, BGP considers various configurable attributes, allowing network engineers to influence the path selection process. Such attributes include weight and local preference. Weight is a Cisco-proprietary attribute that tells a router which local paths are preferred. Local preference is similar to weight, but is specifically designed to tell routers the preferred outbound paths.

Each routing protocol is different. They all use different algorithms, they all have different attributes, and they all consider different information when determining pathways to a destination. To a router, each protocol is like another

language. However, with the creation of these protocols through the 80's and 90's, routers were given the ability to perform route redistribution, translating routes learned from one protocol or source into the local protocol before advertising them to the rest of the network. This feature is essential as it allows various parts of a large network to use the most efficient protocol for its purpose while maintaining communication with the rest of the network. Additionally, it allows a network to be expanded, providing the flexibility to switch routing protocols without reprogramming the old network. This has allowed the Internet to expand to the size it is today. With the control and efficiency of BGP and the ability to translate routing information into different protocols, networks can be configured quickly and easily, and people from all over the globe can communicate in an instant.

Lab Commands

OSPF

```
Router(config)# router ospf <PID>
```

```
Router(config-router)# router-id <RID>
```

Configures the Process ID <PID> and the Router ID <RID> for IPv4.

```
Router(config)# router ospfv3 <PID>
```

```
Router(config-router)# router-id <RID>
```

Configures the Process ID <PID> and the Router ID <RID> for IPv6.

```
Router(config)#int <interface>
```

```
Router(config-if)# ip ospf <PID> area <area>
```

```
Router(config-if)# ipv6 ospf <PID> area <area>
```

Enables an OSPF process on an interface for both IPv4 and IPv6 in a specified area.

EIGRP

```
Router(config)#router eigrp <AS>
```

```
Router(config-router)#network <IP> <wildcard>
```

Tells the router which network addresses <IP> and Wildcard Mask <wildcard> to advertise within the Autonomous System <AS> for IPv4. This enables EIGRP for IPv4.

```
Router(config)#ipv6 router eigrp <AS>
```

```
Router(config-rtr)#eigrp router-id <RID>
```

Tells the router which network addresses <IP> and Wildcard Mask <wildcard> to advertise within the Autonomous System <AS> for IPv6.

```
Router(config)#int <interface>
```

```
Router(config-if)#ipv6 eigrp <AS>
```

Enables EIGRP in the Autonomous System on an interface for IPv6.

IS-IS

```
Router(config)#router isis <area>
```

```
Router(config-router)#net <NET>
```

Enables IS-IS in an Area <area> with a NET Address <NET>.

```
Router(config-router)# is-type level-1
```

Configures the IS-IS type as Level 1: intra-area routing.

```
Router(config-router)#metric-style wide
```

Allows the use of wide metric values, enabling support for advanced IS-IS features.

```
Router(config-router)#log-adjacency-changes
```

A debug command that tells the router to send syslog messages whenever a neighbor goes up or down.

```
Router(config-router)#passive-interface <loopback>
```

Defines the Loopback Interface <loopback> as passive, excluding it from the neighbor discovery process.

```
Router(config)#int <interface>
```

```
Router(config-if)#ip router isis <area>
```

Enables IS-IS on an interface in the Area <area> for IPv4.

```
Router(config)#int <interface>
```

```
Router(config-if)#ipv6 router isis <area>
```

Enables IS-IS on an interface in the Area <area> for IPv6.

BGP

```
Router(config)#router bgp <AS>
```

```
Router(config-router)#bgp router-id <RID>
```

Creates a BGP process on the router in the Autonomous System <AS> with the Router ID <RID>.

```
Router(config-router)#neighbor <IPv4> remote-as <AS>
```

```
Router(config-router)#neighbor <IPv6> remote-as <AS>
```

Defines the BGP neighbor at the IPv4 Address <IPv4> and the IPv6 Address <IPv6>, and the neighbor's Autonomous System <AS>.

```
Router(config-router)#log-neighbor-changes
```

A debug command that tells the router to send syslog messages whenever a neighbor goes up or down.

```
Router(config-router)#address-family ipv4
Router(config-router-af)#network <network> mask <mask>
Router(config-router-af)#neighbor <IPv4> activate
Router(config-router-af)#neighbor <IPv4> next-hop-self
```

Activates the neighbor at the IPv4 Address <IPv4> on the Network <network> with the Subnet Mask <mask> and defines the next hop IP address to be the router's own address.

```
Router(config-router)#address-family ipv6
Router(config-router-af)#network <network>/<mask>
Router(config-router-af)#neighbor <IPv6> activate
Router(config-router-af)#neighbor <IPv6> next-hop-self
```

Activates the neighbor at the IPv6 Address <IPv6> on the Network <network> with the Subnet Mask <mask> and defines the next hop IP address to be the router's own address.

```
Router(config)#route-map <name>
Router(config-route-map)#set local-preference <value>
```

Defines a route-map with the Name <name> and defines the Local Preference Value <value> to influence BGP path selection.

```
Router(config-router)#address-family ipv4
Router(config-router-af)#neighbor <IPv4> route-map <route-map>
Router(config-router-af)#neighbor <IPv4> weight <weight>
```

Controls BGP attributes for a neighbor with the Address <ipv4>, such as weight and local preference, using a Route-Map <route-map> and the weight command to define the value <weight>. Weight cannot be used with IPv6.

Route Redistribution

```
Router(config-router)#redistribute connected
```

Injects connected networks into a routing protocol.

```
Router(config-router)#redistribute <protocol> <area>
```

Injects routes learned via the Protocol <protocol> from the Area or Autonomous System <area> into another routing protocol.

NET Addresses

NET addresses are a unique identifier assigned to an IS-IS router.

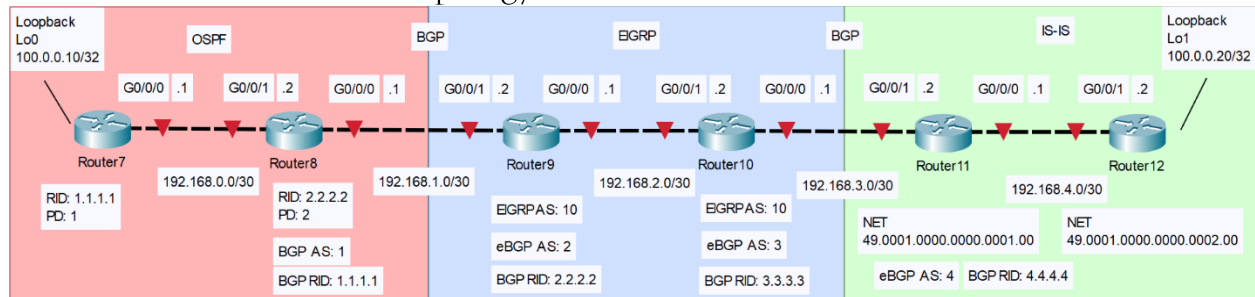
NET = <AFI> <area> <system-id> <NSSEL>

Example: 49.0001.0000.0000.0001.00

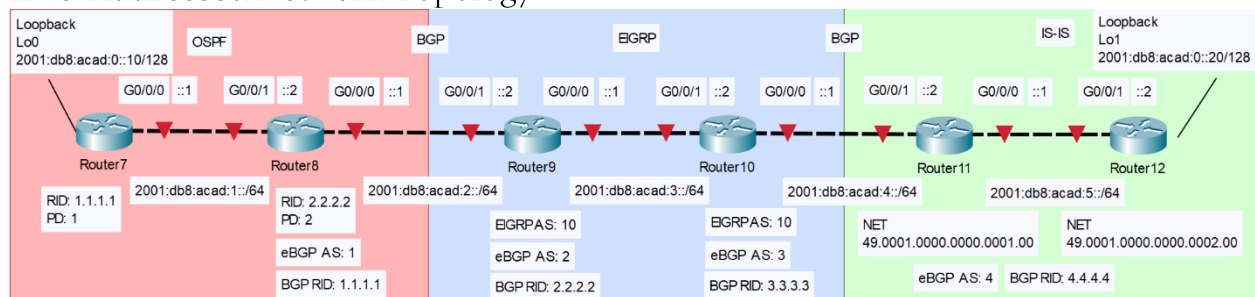
system-id must be unique within the area for Level 1 IS-IS

Network Design

IPv4-Addressed Network Topology



IPv6-Addressed Network Topology



Configurations

Router7

```
version 16.9
hostname Router7
banner motd ^CC UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for OSPF as a part of the eBGP
Cisco Lab ^C
ipv6 unicast-routing
interface Loopback0
 ip address 100.0.0.10 255.255.255.255
 ip ospf 1 area 0
 ipv6 address 2001:DB8:ACAD::10/128
 ipv6 ospf 1 area 0
interface GigabitEthernet0/0/0
 ip address 192.168.0.1 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:DB8:ACAD:1::1/64
 ipv6 ospf 1 area 0
router ospfv3 1
 router-id 1.1.1.1
 address-family ipv6 unicast
 exit-address-family
router ospf 1
 router-id 1.1.1.1
```


Router8

```
version 16.9
hostname Router8
banner motd ^CC UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for OSPF as a part of the eBGP
Cisco Lab ^C
ipv6 unicast-routing
interface GigabitEthernet0/0/0
 ip address 192.168.1.1 255.255.255.252
 ipv6 address 2001:DB8:ACAD:2::1/64
interface GigabitEthernet0/0/1
 ip address 192.168.0.2 255.255.255.252
 ip ospf 2 area 0
 ipv6 address 2001:DB8:ACAD:1::2/64
 ipv6 ospf 2 area 0
router ospfv3 2
 router-id 2.2.2.2
 address-family ipv6 unicast
  redistribute connected route-map BGP-TO-OSPF
  redistribute bgp 1 metric-type 1 route-map BGP-TO-OSPF
 exit-address-family
router ospf 2
 router-id 2.2.2.2
 redistribute connected subnets route-map BGP-TO-OSPF
 redistribute bgp 1 subnets route-map BGP-TO-OSPF
 network 192.168.0.0 0.0.0.3 area 0
router bgp 1
 bgp router-id 1.1.1.1
 bgp log-neighbor-changes
 neighbor 2001:DB8:ACAD:2::2 remote-as 2
 neighbor 192.168.1.2 remote-as 2
 address-family ipv4
  network 192.168.1.0 mask 255.255.255.252
  redistribute ospf 2 route-map OSPF-TO-BGP
  no neighbor 2001:DB8:ACAD:2::2 activate
  neighbor 192.168.1.2 activate
  neighbor 192.168.1.2 next-hop-self
  neighbor 192.168.1.2 weight 100
  neighbor 192.168.1.2 route-map LP in
 exit-address-family
 address-family ipv6
  redistribute ospf 2 route-map OSPF-TO-BGP include-connected
  network 2001:DB8:ACAD:2::/64
  neighbor 2001:DB8:ACAD:2::2 activate
  neighbor 2001:DB8:ACAD:2::2 next-hop-self
  neighbor 2001:DB8:ACAD:2::2 route-map LP in
 exit-address-family
```



```
route-map LP permit 10
  set local-preference 100
route-map BGP-TO-OSPF permit 10
  set metric 50
  set metric-type type-1
route-map OSPF-TO-BGP permit 10
```

Router9

```
version 16.9
hostname Router9
banner motd ^CC UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for EIGRP as a part of the eBGP
Cisco Lab ^C
ipv6 unicast-routing
interface GigabitEthernet0/0/0
  ip address 192.168.2.1 255.255.255.252
  ipv6 address 2001:DB8:ACAD:3::1/64
  ipv6 eigrp 10
interface GigabitEthernet0/0/1
  ip address 192.168.1.2 255.255.255.252
  ipv6 address 2001:DB8:ACAD:2::2/64
router eigrp 10
  network 100.100.0.100 0.0.0.0
  network 192.168.2.0 0.0.0.3
  redistribute bgp 2 route-map BGP-TO-EIGRP
  redistribute connected route-map BGP-TO-EIGRP
router bgp 2
  bgp router-id 2.2.2.2
  bgp log-neighbor-changes
  neighbor 2001:DB8:ACAD:2::1 remote-as 1
  neighbor 192.168.1.1 remote-as 1
  address-family ipv4
    redistribute eigrp 10 route-map EIGRP-TO-BGP
    no neighbor 2001:DB8:ACAD:2::1 activate
    neighbor 192.168.1.1 activate
    neighbor 192.168.1.1 next-hop-self
    neighbor 192.168.1.1 weight 50
    neighbor 192.168.1.1 route-map LP in
  exit-address-family
  address-family ipv6
    redistribute eigrp 10 route-map EIGRP-TO-BGP include-connected
    neighbor 2001:DB8:ACAD:2::1 activate
    neighbor 2001:DB8:ACAD:2::1 next-hop-self
    neighbor 2001:DB8:ACAD:2::1 route-map LP in
  exit-address-family
ipv6 router eigrp 10
  eigrp router-id 3.3.3.3
```

```
redistribute bgp 2 route-map BGP-TO-EIGRP
redistribute connected route-map BGP-TO-EIGRP
route-map BGP-TO-EIGRP permit 10
  set metric 1000000 10 255 1 1500
route-map EIGRP-TO-BGP permit 10
route-map LP permit 10
  set local-preference 50
```

Router10

```
version 16.9
hostname Router10
banner motd ^CC UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for EIGRP as a part of the eBGP
Cisco Lab ^C
ipv6 unicast-routing
interface GigabitEthernet0/0/0
  ip address 192.168.3.1 255.255.255.252
  ipv6 address 2001:DB8:ACAD:4::1/64
interface GigabitEthernet0/0/1
  ip address 192.168.2.2 255.255.255.252
  ipv6 address 2001:DB8:ACAD:3::2/64
  ipv6 eigrp 10
router eigrp 10
  network 192.168.2.0 0.0.0.3
  redistribute bgp 3 route-map BGP-TO-EIGRP
  redistribute connected route-map BGP-TO-EIGRP
router bgp 3
  bgp router-id 3.3.3.3
  bgp log-neighbor-changes
  neighbor 2001:DB8:ACAD:4::2 remote-as 4
  neighbor 192.168.3.2 remote-as 4
  address-family ipv4
    network 192.168.3.0 mask 255.255.255.252
    redistribute connected
    redistribute eigrp 10 route-map EIGRP-TO-BGP
    no neighbor 2001:DB8:ACAD:4::2 activate
    neighbor 192.168.3.2 activate
    neighbor 192.168.3.2 next-hop-self
    neighbor 192.168.3.2 weight 100
    neighbor 192.168.3.2 route-map LP in
  exit-address-family
  address-family ipv6
    redistribute eigrp 10 route-map EIGRP-TO-BGP include-connected
    network 2001:DB8:ACAD:4::/64
    neighbor 2001:DB8:ACAD:4::2 activate
    neighbor 2001:DB8:ACAD:4::2 next-hop-self
    neighbor 2001:DB8:ACAD:4::2 route-map LP in
```

```

    exit-address-family
  ipv6 router eigrp 10
    eigrp router-id 4.4.4.4
    redistribute bgp 3 route-map BGP-TO-EIGRP
    redistribute connected route-map BGP-TO-EIGRP
  route-map BGP-TO-EIGRP permit 10
    set metric 1000000 10 255 1 1500
  route-map EIGRP-TO-BGP permit 10
  route-map LP permit 10
    set local-preference 100

```

Router11

```

version 16.9
hostname Router11
banner motd ^CC UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for IS-IS as a part of the eBGP
Cisco Lab ^C
ipv6 unicast-routing
interface GigabitEthernet0/0/0
  ip address 192.168.4.1 255.255.255.252
  ip router isis 1
  ipv6 address 2001:DB8:ACAD:5::1/64
  ipv6 router isis 1
interface GigabitEthernet0/0/1
  ip address 192.168.3.2 255.255.255.252
  ipv6 address 2001:DB8:ACAD:4::2/64
router isis 1
  net 49.0001.0000.0000.0001.00
  is-type level-1
  metric-style narrow
  log-adjacency-changes
  redistribute connected route-map BGP-TO-IS-IS level-1
  redistribute bgp 4 route-map BGP-TO-IS-IS level-1
  address-family ipv6
    redistribute connected level-1 route-map BGP-TO-IS-IS
    redistribute bgp 4 level-1 route-map BGP-TO-IS-IS
  exit-address-family
router bgp 4
  bgp router-id 4.4.4.4
  bgp log-neighbor-changes
  neighbor 2001:DB8:ACAD:4::1 remote-as 3
  neighbor 192.168.3.1 remote-as 3
  address-family ipv4
    network 192.168.3.0 mask 255.255.255.252
    redistribute connected route-map IS-IS-TO-BGP
    redistribute isis 1 level-1 route-map IS-IS-TO-BGP
  no neighbor 2001:DB8:ACAD:4::1 activate

```

```

neighbor 192.168.3.1 activate
neighbor 192.168.3.1 next-hop-self
neighbor 192.168.3.1 weight 50
neighbor 192.168.3.1 route-map LP in
exit-address-family
address-family ipv6
  redistribute isis 1 level-1 route-map IS-IS-TO-BGP include-
connected
  network 2001:DB8:ACAD:4::/64
  neighbor 2001:DB8:ACAD:4::1 activate
  neighbor 2001:DB8:ACAD:4::1 next-hop-self
exit-address-family
route-map LP permit 10
  set local-preference 50
route-map IS-IS-TO-BGP permit 10
route-map BGP-TO-IS-IS permit 10
  set metric 50
  set metric-type external

```

Router12

```

version 16.9
hostname Router12
banner motd ^CC UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for IS-IS as a part of the eBGP
Cisco Lab ^C
ipv6 unicast-routing
interface Loopback1
  ip address 100.0.0.20 255.255.255.255
  ipv6 address 2001:DB8:ACAD::20/128
interface GigabitEthernet0/0/1
  ip address 192.168.4.2 255.255.255.252
  ip router isis 1
  ipv6 address 2001:DB8:ACAD:5::2/64
  ipv6 router isis 1
router isis 1
  net 49.0001.0000.0000.0002.00
  is-type level-1
  log-adjacency-changes
  passive-interface Loopback1

```

Proof

To verify that these configurations successfully establish a network that utilizes multiple protocols to route traffic and implements additional BGP features, several commands can be used.

Router7#traceroute 100.0.0.20

Type escape sequence to abort.

Tracing the route to 100.0.0.20

VRF info: (vrf in name/id, vrf out name/id)

```
 1 192.168.0.2 1 msec 1 msec 1 msec
 2 192.168.1.2 1 msec 1 msec 0 msec
 3 192.168.2.2 0 msec 1 msec 2 msec
 4 192.168.3.2 1 msec 1 msec 1 msec
 5 192.168.4.2 1 msec 1 msec *
```

Router7#ping 100.0.0.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.0.0.20, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

This command on Router7 shows the success of an IPv4 packet traveling between all of the routers in the line to reach Router12. The 100% success rate demonstrates that IPv4 traffic can traverse the network.

Router12#traceroute 2001:db8:acad:0::10

Type escape sequence to abort.

Tracing the route to 2001:DB8:ACAD::10

```
 1 2001:DB8:ACAD:5::1 1 msec 2 msec 1 msec
 2 2001:DB8:ACAD:4::1 2 msec 2 msec 1 msec
 3 2001:DB8:ACAD:3::1 1 msec 2 msec 1 msec
 4 2001:DB8:ACAD:2::1 2 msec 2 msec 1 msec
 5 2001:DB8:ACAD:1::1 1 msec 2 msec 2 msec
```

Router12#ping 2001:db8:acad:0::10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD::10, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

This command demonstrates the success of an IPv6 packet traveling from Router12 to Router7 and back. This success verifies that IPv6 traffic can traverse the network.

Router8#show ip bgp

BGP table version is 16, local router ID is 1.1.1.1

Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal,

 r RIB-failure, S Stale, m multipath, b backup-path,
f RT-Filter,

 x best-external, a additional-path, c RIB-compressed,
 t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	100.0.0.10/32	192.168.0.1	2		32768	?
*>	100.0.0.20/32	192.168.1.2	5376	100	100	2 ?
*>	192.168.0.0/30	0.0.0.0	0		32768	?
*	192.168.1.0/30	192.168.1.2	0	100	100	2 ?
*>		0.0.0.0	0		32768	i
*>	192.168.2.0/30	192.168.1.2	0	100	100	2 ?
*>	192.168.3.0/30	192.168.1.2	5376	100	100	2 ?
*>	192.168.4.0/30	192.168.1.2	5376	100	100	2 ?

Looking at the result of this command on Router8, we see that each network in the direction from Router8 to Router12 has custom weight and local preference values. This demonstrates the effective implementation of these additional BGP features.

Router7#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

      100.0.0.0/32 is subnetted, 2 subnets
C      100.0.0.10 is directly connected, Loopback0
O E1      100.0.0.20 [110/51] via 192.168.0.2, 00:17:04,
GigabitEthernet0/0/0
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.0.0/30 is directly connected,
GigabitEthernet0/0/0
L      192.168.0.1/32 is directly connected,
GigabitEthernet0/0/0
      192.168.1.0/30 is subnetted, 1 subnets

```

```

O E1          192.168.1.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0
    192.168.2.0/30 is subnetted, 1 subnets
O E1          192.168.2.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0
    192.168.3.0/30 is subnetted, 1 subnets
O E1          192.168.3.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0
    192.168.4.0/30 is subnetted, 1 subnets
O E1          192.168.4.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0

```

This command output shows that Router7 has OSPF routes that have been learned from external networks. These other networks were configured with different protocols like IS-IS and EIGRP, meaning these had to be translated to OSPF before Router7 could add them into its routing table. Therefore, translating between protocols is functioning.

Router7#show ipv6 route

```

IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
    B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
    IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX -
EIGRP external
    ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -
Redirect
    O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2
    ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, la - LISP alt
lr - LISP site-registrations, ld - LISP dyn-eid, lA - LISP
away
    le - LISP extranet-policy, a - Application
LC 2001:DB8:ACAD::10/128 [0/0]
    via Loopback0, receive
OE1 2001:DB8:ACAD::20/128 [110/51]
    via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
C 2001:DB8:ACAD:1::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
OE1 2001:DB8:ACAD:2::/64 [110/51]
    via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
OE1 2001:DB8:ACAD:3::/64 [110/51]
    via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
OE1 2001:DB8:ACAD:4::/64 [110/51]
    via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0

```



```

OE1 2001:DB8:ACAD:5::/64 [110/51]
    via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
    via Null0, receive

```

This command shows the same successful protocol translation for IPv6. These external OSPF routes come from networks that are routing with IS-IS and EIGRP. To get into Router7's routing table, they had be converted into BGP routes that were then distributed into OSPF.

Router8#show ip route

```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile,
B - BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-
IS level-2
      ia - IS-IS inter area, * - candidate default, U - per-user
static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l
- LISP
      a - application route
      + - replicated route, % - next hop override, p - overrides
from PfR

```

Gateway of last resort is not set

```

      100.0.0.0/32 is subnetted, 2 subnets
O       100.0.0.10 [110/2] via 192.168.0.1, 00:22:28,
GigabitEthernet0/0/1
B       100.0.0.20 [20/5376] via 192.168.1.2, 00:22:02
      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/30 is directly connected,
GigabitEthernet0/0/1
L       192.168.0.2/32 is directly connected,
GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/30 is directly connected,
GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected,
GigabitEthernet0/0/0
      192.168.2.0/30 is subnetted, 1 subnets
B       192.168.2.0 [20/0] via 192.168.1.2, 00:30:53
      192.168.3.0/30 is subnetted, 1 subnets

```

```
B          192.168.3.0 [20/5376] via 192.168.1.2, 00:30:53
          192.168.4.0/30 is subnetted, 1 subnets
B          192.168.4.0 [20/5376] via 192.168.1.2, 00:22:32
```

This command verifies the protocol translation and route distribution from another angle. Router8 has learned routes from all three protocols – OSPF, EIGRP, and IS-IS – and has adapted them into a BGP network.

Problems

After the first day of work, we returned the next day and uploaded our configurations to the routers. However, checking the routing tables showed that adjacencies failed. We checked the interfaces and saw that they were all administratively down despite the “no shut” command being listed for each configured interface. Re-entering the command flipped the ports to an up state, and the route tables filled with IP addresses.

After getting OSPF and EIGRP configured with BGP between these two networks, we struggled to get redistribution working. We verified that IP addresses were configured correctly and that each individual routing protocol was working. However, we had no luck in fixing the redistribution. So, we researched for some support and found a senior network engineer on the Cisco Community forums who explained how to route between OSPF and EIGRP with BGP as the middleman. This explanation cleared up the purpose of route-maps and how to use the redistribute command. With his help, we saw BGP routes being established and redistributed to devices on different networks.

However, after we thought we had figured out the IPv4 configuration, we noticed that we could not ping the 192.168.2.0/24 EIGRP network. We could ping across it to other networks, but not to the interfaces directly. After some digging through the configurations, we realized that the redistribution commands did not distribute directly connected neighbors. After adding the redistribute connected command, the pings worked. With that, IPv4 configurations were done for OSPF and EIGRP.

When working with IS-IS, we used the same commands, but the pings failed. It took some research to learn that the redistribution commands default to IS-IS level-2, whereas our IS-IS network used level-1. After manually assigning level-2, everything worked. All that was left was configuring IPv6.

In theory, implementing IPv6 redistribution is an easy process, as it uses very similar commands to IPv4. However, our pings initially failed. Unfortunately, we had forgotten to add the same redistribute connected command that we had forgotten earlier. After catching this mistake and correcting our configurations, we were able to ping anywhere across our networks with both IPv4 and IPv6.

Conclusion

There is a wide variety of routing protocols that can be implemented in a network. However, with its scalability, security, and precise control, BGP stands out among the rest. As the routing protocol for the Internet, it is important to understand how it works and how it can be manipulated within a network. To run in parallel with other routing protocols, routes can be redistributed, injecting them into the local protocol for that network. With this, all networks can communicate with one another quickly and securely, and complex networks can future-proof their design to ensure reliability for years to come.